

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		5.81×10^{-4} or 5.8×10^{-4} = 2 marks (accept any d.ps) If incorrect award 1 mark for 18/3099100 x100 0.00058081 5.80×10^{-4}		2		2	2	
	(b)	(i)		Gram – have capsule/lipopolysaccharide (1) Bacillus rod shaped (1)	2			2		
	(c)	(i)		172000 = 2 marks Accept standard form If incorrect award 1 mark for 86000× 2 86000 (no dilution factor)		2		2	2	
		(ii)	I	there were too many <u>colonies</u> to count/ <u>colonies</u> {merged/clumping}			1	1		1
			II	The extra dilution gives additional error/may not have mixed fully/inaccurate representation of whole sample/not valid to count less than 30 colonies/too few to be statistically significant			1	1		1
	(d)			azithromycin (no mark for name only) need to use less antibiotic/use of data e.g. Anti A only needs 10% to kill all bacteria. C needs 30% to kill all bacteria(1) so cheaper for government/less likely to lead to antibiotic resistance (1)			2	2		
				Question 2 total	2	4	4	10	4	2